

PROJECT AND TEAM INFORMATION

Project Title

(Try to choose a catchy title. Max 20 words).

Hospital Management System with Emergency Patient Priority Queue

Student / Team Information

Team Name: Team #:		Mentor:	Mr. RAUNAK KUMAR Department of Computer Applications
Team member 1 (Team Lead) (Name, student ID / university roll no., semester & section)	PRANJAL KUMAR ID / Roll No.: 2510320140 / 2105749 BCA 3rd Sem, Section G1	Photo	
Team member 2 (Name, student ID / university roll no., semester & section)	SARANSH TOMAR ID / Roll No.: 2510320091 / 2105795 BCA 3rd Sem, Section G4	Photo	
Team member 3 (Name, student ID / university roll no., semester & section)	VINITA ID / Roll No.: 2510320092 / 2105868 BCA 3rd Sem, Section G1	Photo	

PROPOSAL DESCRIPTION (10 pts)

Motivation (1 pt)

(Describe the problem you want to solve and why it is important. Max 300 words).

Small and medium-sized clinics often manage patient registration, bed assignment, appointments, prescriptions and billing through paper registers or basic spreadsheets. These methods can cause long waiting times, misplaced records, difficulty checking bed availability, prescription safety issues and billing errors. The proposed Hospital Management System addresses these problems through a desktop application developed using Python and SQLite. It stores patient information, bed details, doctor records and billing data in one place. Its key feature is an Emergency Patient Priority Queue, which ensures that critical patients are placed ahead of less serious cases. Priority is determined using information such as oxygen level, pulse rate and severity. This project is important because it demonstrates how programming, database management, object-oriented concepts and data structures can be applied to improve healthcare administration.

State of the Art / Current Solution (1 pt)

(Describe how the problem is solved today (if it is). Max 200 words).

Many small clinics currently depend on handwritten registers, spreadsheets, telephone communication and manual calculations. Patient records are difficult to search, bed occupancy may not be visible in real time, and emergency patients may be handled using a basic first-come, first-served process. Doctors may also need to manually search paper records for allergy information. Billing is frequently calculated with a calculator, increasing the possibility of mistakes. Although some hospital software products provide registration, appointments or billing modules, they may be costly or complex for small clinics. The proposed system combines essential functions in a simple, local desktop application.

Project Goals and Milestones (2 pts)

(Describe the project general goals. Include initial milestones and other milestones. Max 300 words).

The main goal is to develop a user-friendly desktop Hospital Management System using Python, Tkinter and SQLite. The objectives are to: (1) register and search patient records; (2) maintain doctor and appointment information; (3) display General, ICU and Oxygen bed availability; (4) implement an Emergency Patient Priority Queue using severity and vital signs; (5) store patient allergies and provide prescription allergy alerts; (6) automate billing for doctor fees, bed charges and medicines; (7) apply object-oriented programming concepts such as classes, objects, inheritance and encapsulation; and (8) test the system using valid, invalid and edge-case data. Milestones include requirement analysis, database design, interface development, core module implementation, integration, testing and documentation.

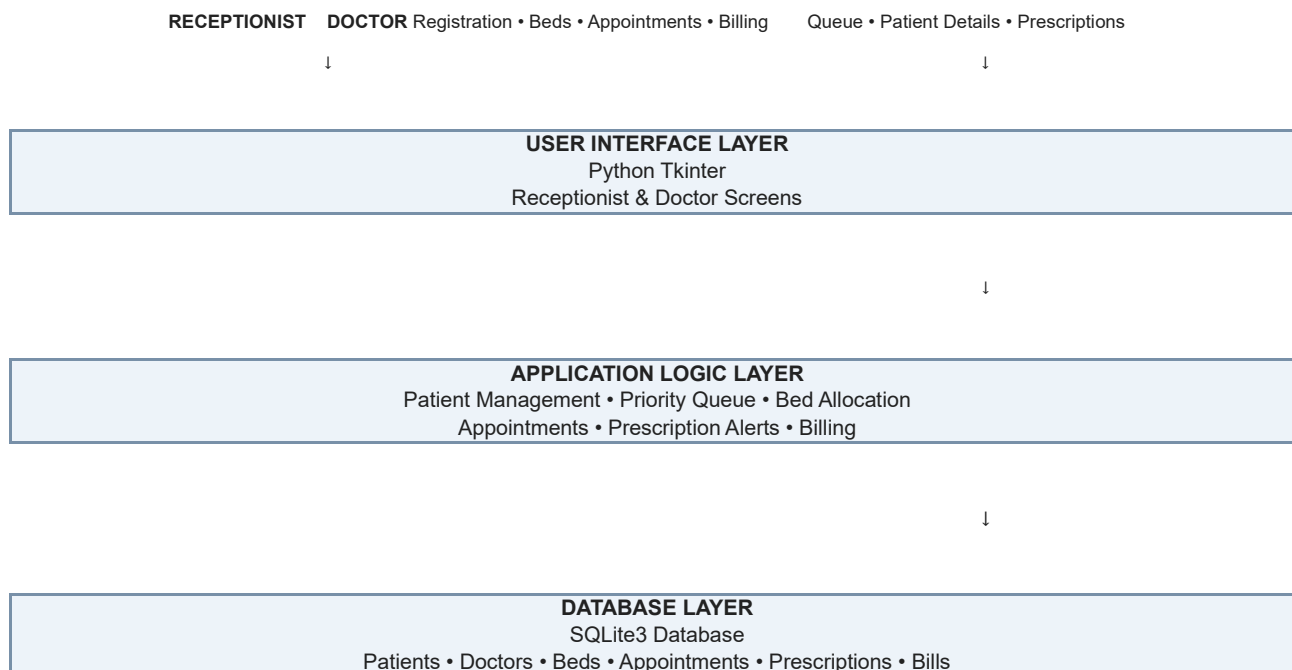
Project Approach (3 pts)

(Describe how you plan to design the solution, including platforms and technologies. Max 300 words).

The application will follow a three-tier architecture. The user-interface layer will be developed with Python Tkinter and will provide screens for receptionists and doctors. The application-logic layer will contain Python classes and methods for patient registration, appointments, priority queue operations, bed allocation, prescription checks and bill calculation. The database layer will use SQLite3 to store patients, doctors, beds, appointments, prescriptions and bills. A priority queue will rank emergency cases so that patients with severe conditions receive earlier attention. During prescription entry, the system will compare selected medicines with stored allergy information and display a warning when a possible conflict is detected. Development will be incremental: each module will be implemented, tested with sample records, refined and then integrated. The tools include Python 3, Tkinter, SQLite3 and an IDE such as PyCharm or VS Code.

System Architecture (High Level Diagram) (2 pts)

(Provide an overview of the system, identifying its main components and interfaces in the form of a diagram using a tool of your choice).



Project Outcome / Deliverables (1 pt)

(Describe what are the outcomes / deliverables of the project. Max 200 words).

The expected outcome is a functional offline desktop Hospital Management System that centralizes patient and clinic records. The deliverables include patient registration and search, doctor and appointment management, bed occupancy display, emergency patient priority queue, prescription allergy alert, automated billing, SQLite database storage, Tkinter interfaces for receptionists and doctors, source code, test cases, screenshots and project documentation. The project will demonstrate the practical use of Python, SQLite, object-oriented programming and the priority queue data structure.

Assumptions

(Describe the assumptions, if any, you are making to solve the problem. Max 100 words).

The system is intended for small or medium-sized clinics and will initially run on a local computer without an internet connection. Users are assumed to enter correct patient, allergy, bed and billing information. Emergency priority is based on the recorded severity and vital signs and is intended as an academic software feature, not a replacement for professional medical judgment.

References

(Provide a list of resources or references utilised for the completion of this deliverable. You may provide links).

1. Python Official Documentation – Tkinter and sqlite3: <https://docs.python.org/3/>
2. Reema Thareja, Data Structures Using C, Oxford University Press.
3. E. Balagurusamy, Object-Oriented Programming Concepts.
4. GeeksforGeeks – Priority Queue and Heap Implementation Tutorials.
5. SQLite Official Documentation: <https://www.sqlite.org/docs.html>